

# Nathan Norman

## Senior Software Engineer | Agent Platforms & Backend

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Backend and platform engineer with 11 years of experience building production systems. Recent work spans AI agents, MCP integrations, Claude Code plugins, and AWS infrastructure.

### EXPERIENCE

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#### Senior Software Engineer, Data Platform

September 2021 – September 2026

Toast · Remote

Backend engineer on the analytics platform behind restaurant reporting (~45 developers contribute monthly). Shipped production agents and MCP integrations for platform Q&A and ops workflows, plus migrations and ECS capacity work on Kafka, Spark, Glue, and AWS.

- Built and shipped apex-researcher, an agent service (Kotlin, Koog, A2A protocol) on ECS that answers questions about Toast's analytics platform using DataHub and Sourcegraph through MCP.
- Built an investigation agent for Red Flag support escalations that works from triage through evidence gathering to a written report, with its own validators, tools, and model for each phase. Rewrote the original Python orchestration as a Kotlin/Koog workflow with an MCP tool sidecar for Splunk, Datadog, Sourcegraph, and GitHub.
- Built a ticket router that predicts the owning team for support escalations, running in shadow mode and scoring itself against final assignments before it is trusted to reassign anything.
- Built Slack agents for production alert triage and ticket investigation using a Postgres catalog assembled from 3,000+ alerts.
- Exposed internal systems (data catalog, notebook execution, CI/CD, log search) to any agent through MCP servers with standardized SSO auth.
- Created toast-bedrock-adapter, Toast's standard LLM access layer: an Anthropic- and OpenAI-compatible proxy to Amazon Bedrock with per-user cost attribution and OpenTelemetry metrics. Other teams adopted it on their own and contributed PRs.
- Published Claude Code plugins and skills to Toast's internal marketplace, connecting agents to logs, code search, CI/CD, deploys, project tracking, and Slack. Engineers across the org adopted them.
- Turned a multi-day engagement with a senior engineer into a self-serve workflow with a pipeline-authoring skill that generates Kafka-to-Spark config and SQL and validates it locally.
- Ran weekly AI Enablement Office Hours covering Claude Code plugins, hooks, session memory, live incidents, and running agents in production.
- Led the Labor Druid-to-OpenSearch migration, including the local stack playbook, the labor-olap-consumer service, and cutover runbooks.
- Built target-tracking autoscaling for analytics services on ECS, and found about \$120K/year in recoverable storage by analyzing roughly 3 million S3 access-log entries.

**Key tech** Python, Java, Kotlin, Koog, MCP, Claude Code, Bedrock, AWS (ECS, Lambda, S3, DynamoDB, Glue, EventBridge), Kafka, Spark, OpenSearch, Datadog, Splunk, PostgreSQL, Terraform

#### Senior Software Engineer

July 2018 – September 2021

Bandwidth · Raleigh, NC

- Led reporting and notifications for Bandwidth Dashboard: assigned work, unblocked engineers, reviewed team code, and represented incidents at the company Outage Review Board.
- Led Dashboard geo redundancy across two datacenters, including infrastructure specifications, Ansible failover, Route 53 DNS automation, replicated S3 storage, data migration, and cross-team failover tests.

- Administered SonarQube for roughly 120 engineers, managing access and helping teams add static analysis to their builds.

**Key tech** Java, Spring Boot, AWS, Ansible, GitHub Actions, Datadog, SonarQube, CloudFormation, Route 53

## Software Engineer II

May 2015 – July 2018

Oracle (Bronto) · Durham, NC

- Replaced a data processing system handling 4–5 billion records per day.
- Built batch jobs to inspect, clean, and delete 70+ billion stale records from production databases.

**Key tech** Java, Hadoop, MapReduce, MySQL

## PROJECTS

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**review-fix-agent.** A local Koog service that reviews a branch before each push, running security, architecture, code quality, testing, and standards checks in parallel.

**toast-archiving review agent.** A command-line PR reviewer for archiving and schema changes. It reviews the exact commits in the PR and skips Docker and Gradle builds when only config changed.

## SKILLS

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**Languages and frameworks:** Python, Java, Kotlin, SQL, Spring Boot

**AI agents:** Claude Code, MCP, Koog, A2A protocol, Amazon Bedrock

**AWS:** ECS, Lambda, S3, DynamoDB, Glue, EventBridge, Route 53, CloudFormation

**Data:** Kafka, Spark, PostgreSQL, MySQL, Hadoop

**Infrastructure and CI:** Terraform, Ansible, Docker, Gradle, GitHub Actions, SonarQube

**Observability:** Datadog, Splunk, FireHydrant

## EDUCATION

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**B.S. Software Engineering and Databases,** Certificate in Information Security, Radford University, 2015

**B.S. Human Nutrition, Foods and Exercise,** Virginia Tech, 2010

## AWARDS

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**MVP Developer of the Year, 2020.** Peer-voted, Bandwidth Engineering (120+ engineers)

**Code Reviewer of the Year, 2nd Place, 2019.** Peer-voted, Bandwidth Engineering